

Collective Robotics

Part 5: Modeling

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RECAP

**What did
you learn
last time?**





IT'S MEME

TIME

imgflip.com

3 minutes of thinking

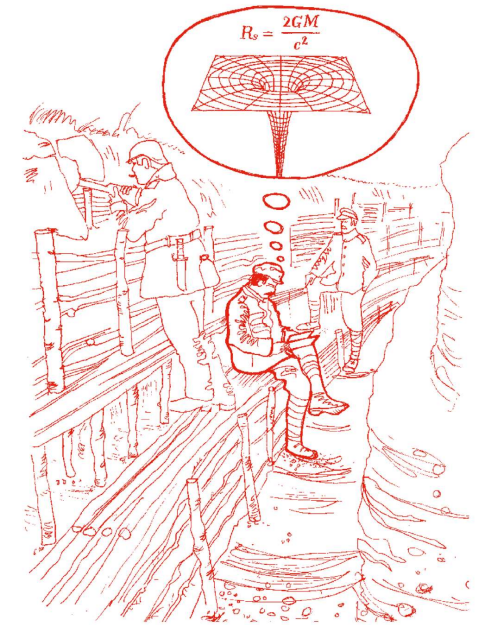
- Why we do modeling?
- How to model swarm?
- Which challenges do you see in modeling swarm systems?

Modeling

Check the Slides in Lea

Modeling as essential part of the scientific approach

- abstraction: parts of the modeled system are omitted intentionally
- simplification: the modeled system is transferred into a simpler system which is easier to understand
- formalization: the modeled system is described by a formal system that allows the derivation of new insights by means of logic and mathematics



Karl Schwarzschild (1916)
<https://spark.iop.org/schwarzschilds-war>

Why models of swarm robotic systems? (1/2)

many entities $\rightarrow$ high dimension

For example, in 2-dimensional space and swarm size N

each agent $i \in \{1, 2, \dots, N\}$ has a . . .

- position $x_i = (x_i^1, x_i^2)$
- velocity $v_i = (v_i^1, v_i^2)$
- state s_i

a configuration is given by

$$\gamma = (x_1, x_2, \dots, x_N, v_1, v_2, \dots, v_N, s_1, s_2, \dots, s_N)$$

configuration space Γ ($\gamma \in \Gamma$)

has dimension: $\dim(\Gamma) = 2N + 2N + N = 5N$

wanted: mapping $f : \Gamma \rightarrow \Phi$ *with* $\dim(\Phi) \ll \dim(\Gamma)$

+ certain properties

Why models of swarm robotic systems? (2/2)

The main objective of modeling in swarm robotics is dimension reduction.

We abstract away microscopic details in order to detect key features and general principles of operation.

Could also be done automatically in a brute-force approach with methods from machine learning and statistics

(e.g., principal component analysis, feature extraction, etc.)

⇒ but non-constructive

(no detection of general principles possible)

What do we want to model?

The key feature of the modeled system has to be representable in the model's configuration space Φ .

temporal course of the system:

- time-dependent model
- discrete or continuous time?

and/or

positions of agents or agent densities:

- spatial model
- discrete or continuous?

Requirements – time-dependent model

Evolution of the swarm robotic system in terms of configurations γ (assuming discrete time for simplicity):

$$\gamma_t, \gamma_{t+1}, \gamma_{t+2}, \gamma_{t+3}, \dots$$

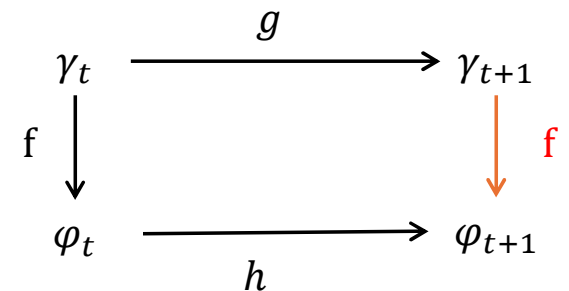
Update rule as mapping $g(\gamma_t) = \gamma_{t+1}$

Similarly in our model: $h(\varphi_t) = \varphi_{t+1},, \text{ for } \varphi_t, \varphi_{t+1} \in \Phi$

the temporal course of γ_t and φ_t should ideally correspond:

$$h(f(\gamma_t)) \stackrel{!}{=} f(g(\gamma_t))$$

for mapping f implementing the modeling abstractions as mentioned before



Example 1 – Binary, collective decision

Collective decision-making system,

binary decision (2 options: A or B),

2-d spatial system

agent state $s_i \in \{A, B\}$

dimension of configuration space: $\dim(\Gamma) = 2N + 2N + N = 5N$

simple, time-dependent model: $f(\gamma) = \frac{|\{s_i | s_i=A\}|}{N} = \frac{[\text{number of agents in favor of option A}]}{N} = \varphi$

φ describes the key feature, which is the decision-making process

Dimension of the model: $\dim(\Phi) = 1$ (just 1 state variable!)

How does the update rule ($h(\varphi_t) = \varphi_{t+1}$) of the model look like?

What about a collective decision-making system with 3 options (A, B, C)?

Example 2 – Rate equations

say an agent i can be in one of several states $s_i \in S$ with $S = \{0, 1, 2, 3, \dots\}$

Approach: time-dependent model, one *state variable* w_j for each state j that gives the swarm fraction currently in state j

$$w_j(t) = \frac{|\{s_i(t) | s_i(t) = j\}|}{N}, \text{ for a given time } t$$

Note: $\sum w_i = 1$

update rule as ordinary differential equation (ODE): $\frac{dw_i}{dt} = \dots$

→ depends on probabilities of state transitions

dimension of the model: $\dim(\Phi) = |S|$

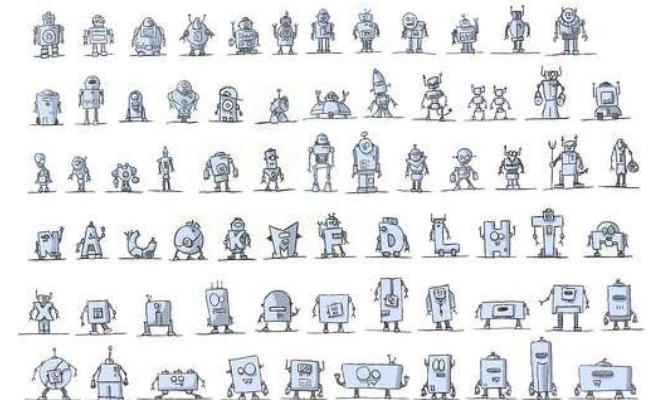
(could even be simplified to $\dim(\Phi) = |S| - 1$ due to $\sum w_i = 1$)

Micro-macro

2 levels: micro and macro

The **microscopic level** (or **micro-level**) of a swarm system is the individual level. In a microscopic description of a swarm system the individual agent is addressed, that is, we can determine, for example, trajectories of agents and distances between agents.

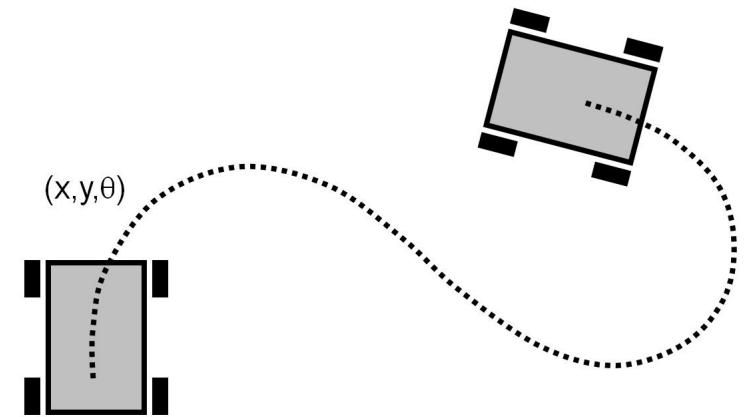
The **macroscopic level** (or **macro-level**) of a swarm system is the group level. Individual agents are not addressed as we work only on group fractions or densities of agents, that is, trajectories of individual agents cannot be determined, only, for example, a mean trajectory of the swarm's barycenter.



Micro and macro

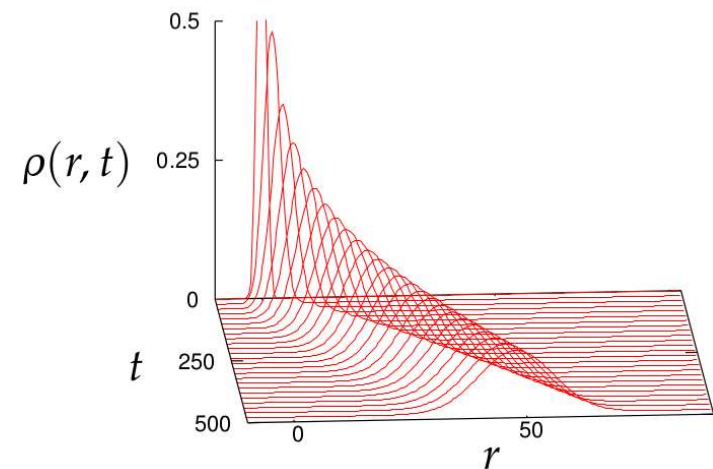
Micro-level:

- each agent modeled with position, heading, . . .
- agent–agent interactions directly modeled



Macro-level:

- only agent density
- no concept of trajectories anymore
- modeling agent–agent interactions difficult



Potential for insight vs microscopic details

To gain insight into the interplay between microscopic interactions and macroscopic features, it is important to find a level of description that, on the one hand, considers specific features of the system and is suitable for reflecting the origination of new qualities, but, on the other hand, is not flooded with microscopic details. In this respect, agent models have become a very promising tool mainly for simulating complex systems. A commonly accepted theory of agent systems that also allows analytical investigations is, however, still pending because of the diversity of the various models invented for particular applications.

Schweitzer (2003)

Micro-macro problem (1/3)



- micro-macro link or micro-macro problem
- technical term in sociology (Schillo et al., 2000; Alexander et al., 1987; Schelling, 1978)
- micro-interactions: interactions between individual humans, for example, actions that are aimed to change the actions of others
- macro-structures: structural rules defined, for example, by the human society
- micro-macro link: mutual influence of the macro-structure to micro-interactions and vice versa

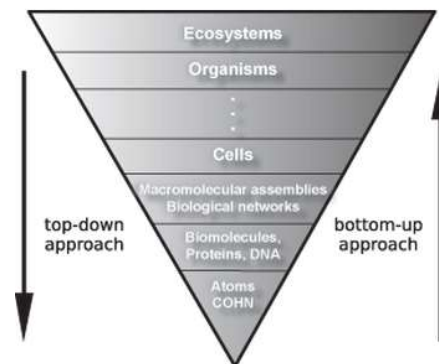
Micro-macro problem (2/3)

The action of an individual might be influenced or caused by the individual's perception of the **macro-structure**. This behavior influences, in turn, the macro-structure leading to a **closed loop**. Due to this loop structure it is difficult to distinguish whether the **cause** of a considered action is found on the micro- or the macro-level as indicated by the term “micro-macro problem.”

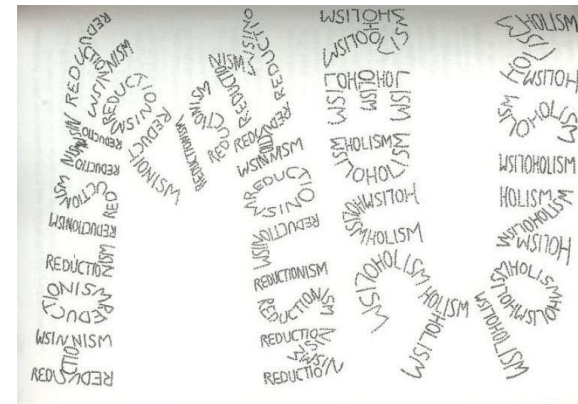


Micro-macro problem (3/3)

- problem to distinguish cause and effect due to closed loop
- reductionism (bottom-up): explain a system by reducing the system into its constituents
- holism (top-down): assumption that not everything is determined by the lower levels, hence allow 'high-level' explanations



Reductionism vs Holism



Reductionism

REDUCTIONISM REDUX

Steven Weinberg

NATURE'S IMAGINATION:
THE FRONTIERS OF SCIENTIFIC VISION
edited by John Cornwell,
introduction by Freeman Dyson,
Oxford University Press,
212 pp., 23.00USD

It used to be traditional for college courses on the history of philosophy to begin around 600 BC with Thales of Miletus. According to later writers, Thales taught that everything is made of water. Learning about Thales, undergraduates had the healthy experience of starting their study of philosophy with a doctrine that they knew to be false.

- Freeman Dyson on reductionism in physics: “to reduce the world of physical phenomena to a finite set of fundamental equations”
- scientific findings are all connected and build up on each other, they originate from the same origin (set of fundamental equations)

(Weinberg, 1995)

Holism

4 August 1972, Volume 177, Number 4047

SCIENCE

More Is Different

Broken symmetry and the nature of
the hierarchical structure of science.

P. W. Anderson

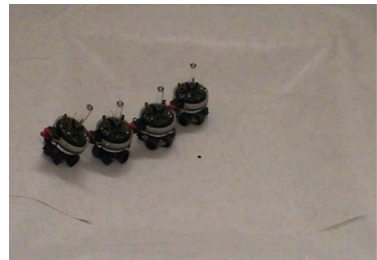
less relevance they seem to have to the very real problems of the rest of science, much less to those of society.

The constructionist hypothesis breaks down when confronted with the twin difficulties of scale and complexity. The behavior of large and complex aggregates of elementary particles, it turns out, is not to be understood in terms of a simple extrapolation of the properties of a few particles. Instead, at each level of complexity entirely new properties appear, and the understanding of the new behaviors requires re-

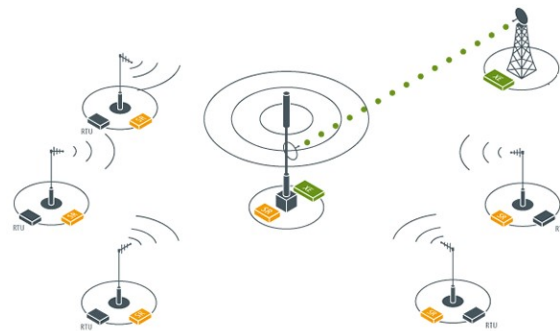
- “At each stage, entirely new laws, concepts and generalizations are necessary, requiring inspiration and creativity to just as great a degree as in the previous one. Psychology is not applied biology nor is biology applied chemistry.”
- the research program of reductionism is not the solution for everything

From micro to macro and back (1/3)

How to model structure formation with densities?



How to model point2point communication etc.?



From micro to macro and back (2/3)

micro $\rightarrow$ macro:

say the swarm should be modeled as swarm density

say 2 robots stop when approaching each other

How to model such agent–agent interactions based on densities?

$\Rightarrow$ Similar to a reaction–diffusion system:

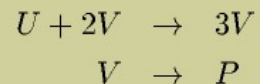
calculate probabilities for interactions (reactions) and account for the respective effects.

Gray Scott Model of Reaction Diffusion

Equations:

$$\begin{aligned}\frac{\partial u}{\partial t} &= r_u \nabla^2 u - uv^2 + f(1 - u) \\ \frac{\partial v}{\partial t} &= r_v \nabla^2 v + uv^2 - (f + k)v\end{aligned}$$

Chemical Reaction:



U and V and P are chemical species.

u and v represent their concentrations.

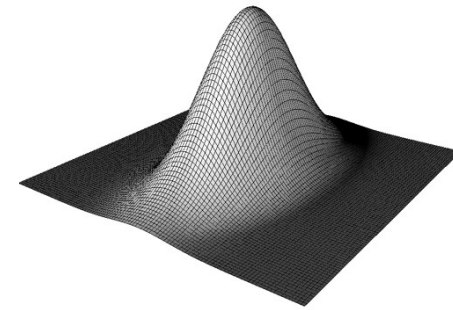
r_u and r_v are their diffusion rates. k represents the rate of conversion of V to P .

f represents the rate of the process that feeds U & drains U, V, P .

<http://groups.csail.mit.edu/mac/projects/amorphous/GrayScott/>



From micro to macro and back (3/3)



macro $\rightarrow$ micro:

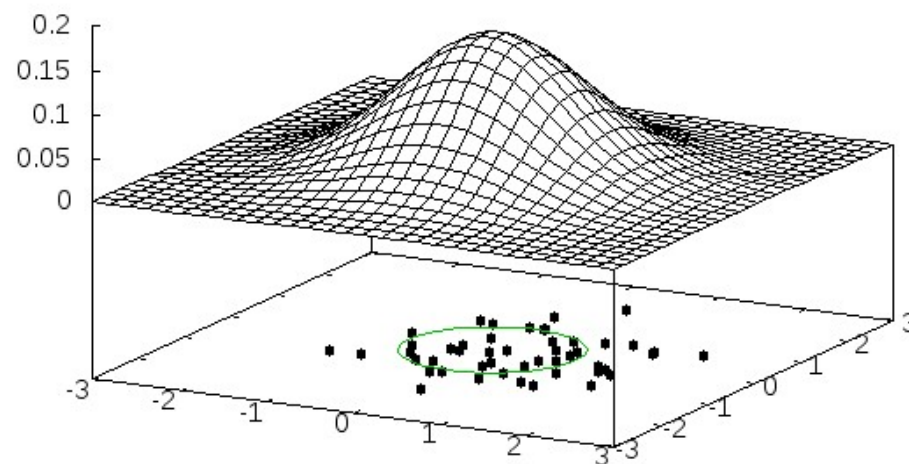
Say the swarm is modeled by swarm density

How to get a microscopic system state out of this macroscopic system state?

Sampling from the distribution gives us realizations of this distribution.

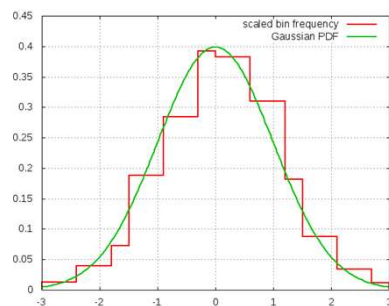
Sampling from a distribution (1/3)

50 random samples from a 2D Gaussian PDF with unit variance, zero mean and no dependence

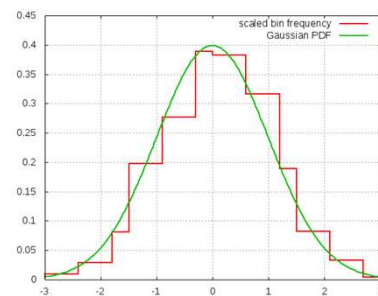


The realization (sampled robot positions) has different qualities compared to the distribution (e.g., distances).

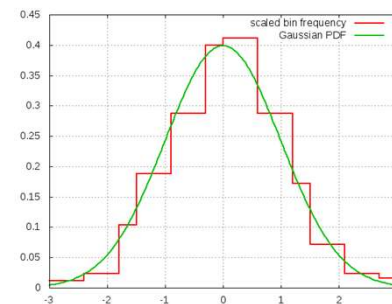
Sampling from a distribution (2/3)



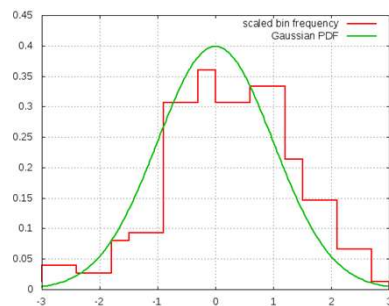
5000 samples



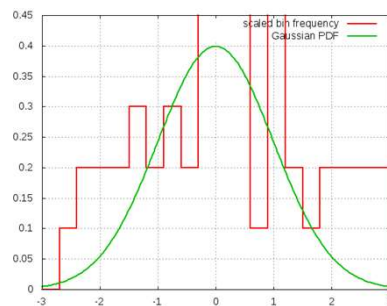
1500 samples



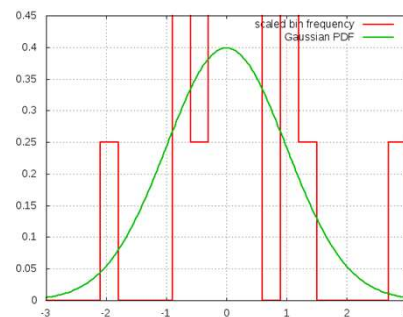
500 samples



150 samples

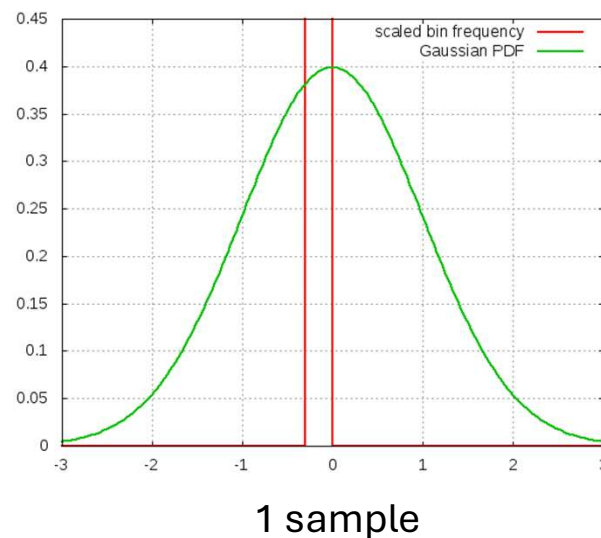


50 samples



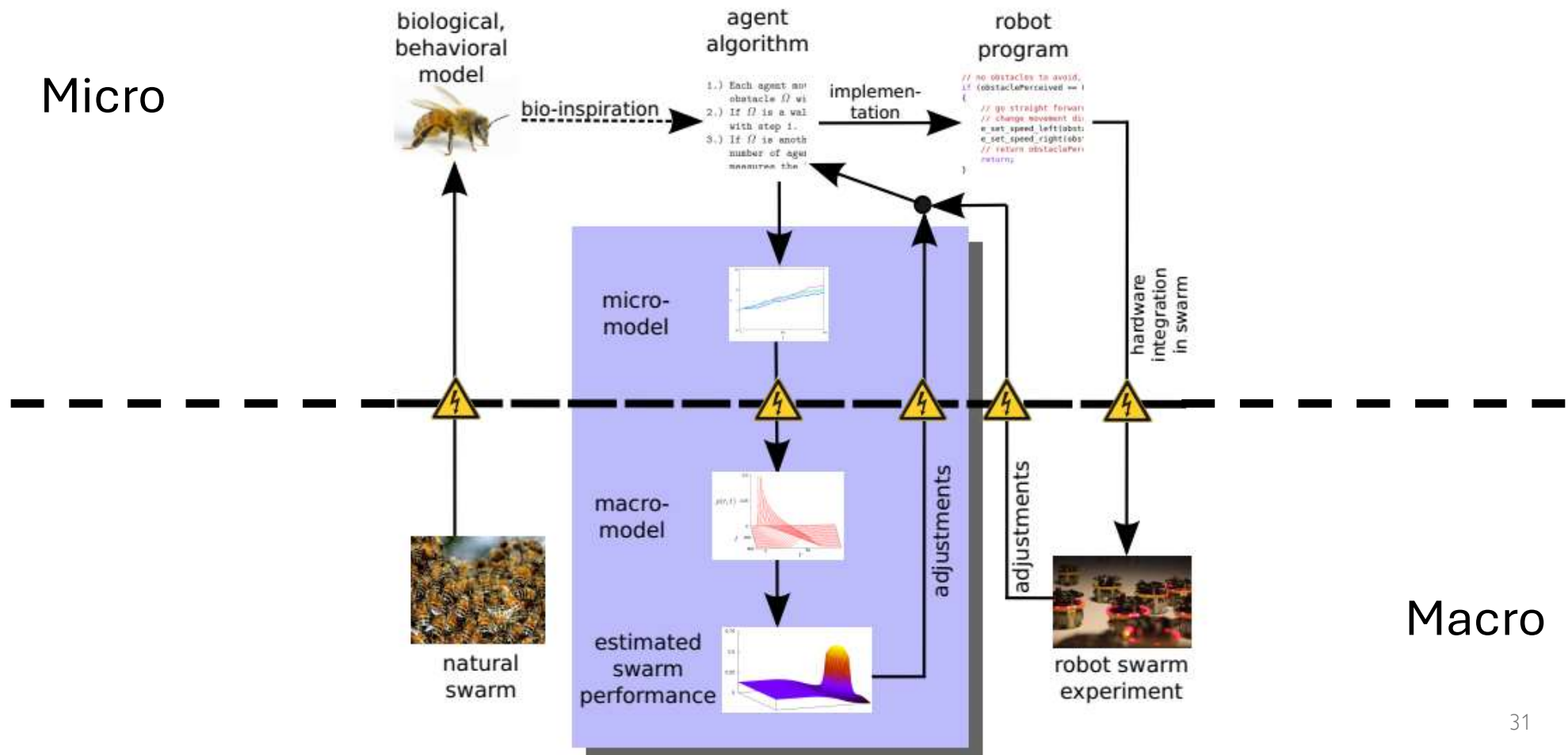
20 samples

Sampling from a distribution (3/3)



Also note that even a single sample can be mathematically modeled as a distribution by the Dirac delta function.

Micro-macro models



Grand Challenge of Swarm Robotics

- Find a general and scientifically sound conversion of swarm behaviors to behaviors of single agents(and the other way around).
- Can this be done (semi-)automatically?

“holy grail” of swarm robotics:

- Construct a **macro-micro compiler** with a ‘swarm language’
 - input: description of swarm behavior in swarm language
 - output: code for single agent

“Interim” approaches

Micro-macro compiler not yet available,

Hence we need interim approaches to the micro-macro problem

Candidate solutions are:

- machine learning (multi-agent learning)
- artificial evolution of robot controllers(Thats another lecture)
- human in the loop supported by **models**

Modeling techniques

Time-dependent models

- ordinary differential equations (ODE)
- recurrence relations (difference equations)
- stochastic ODE

Models representing some spatial features

- graph theory (connected components, node degree, random graphs, etc.)
- networks (routing, multi-hop communication, etc.)

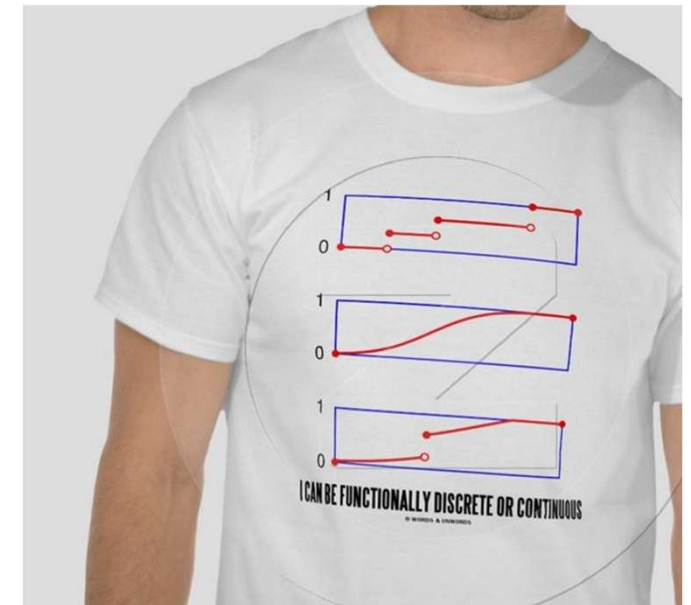
fully spatial models

- partial differential equations (PDE)
- simulation (grid worlds, floating point positions)

'FAQ': Discrete or continuous?

Practical approach:

- many of the models cannot be solved analytically, hence they have to be discretized anyways.
- However, a continuous notation often is concise and easy to handle.



Rate equations

The rate equations approach

a simple rate equation: $r = kAB$

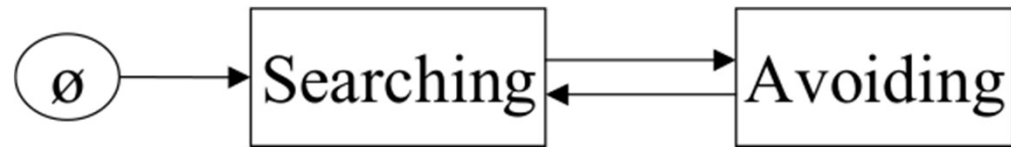
- chemical reaction between two reactants
- concentrations of the chemical species are A and B
- k is the rate coefficient

Say we have $A + B \rightarrow C$ then an ordinary differential equation (ODE) is defined by the reaction rate:

$$\frac{dC}{dt} = kAB,$$

for concentrations A, B, and C
(law of mass action)

Rate equations of swarm robotic systems



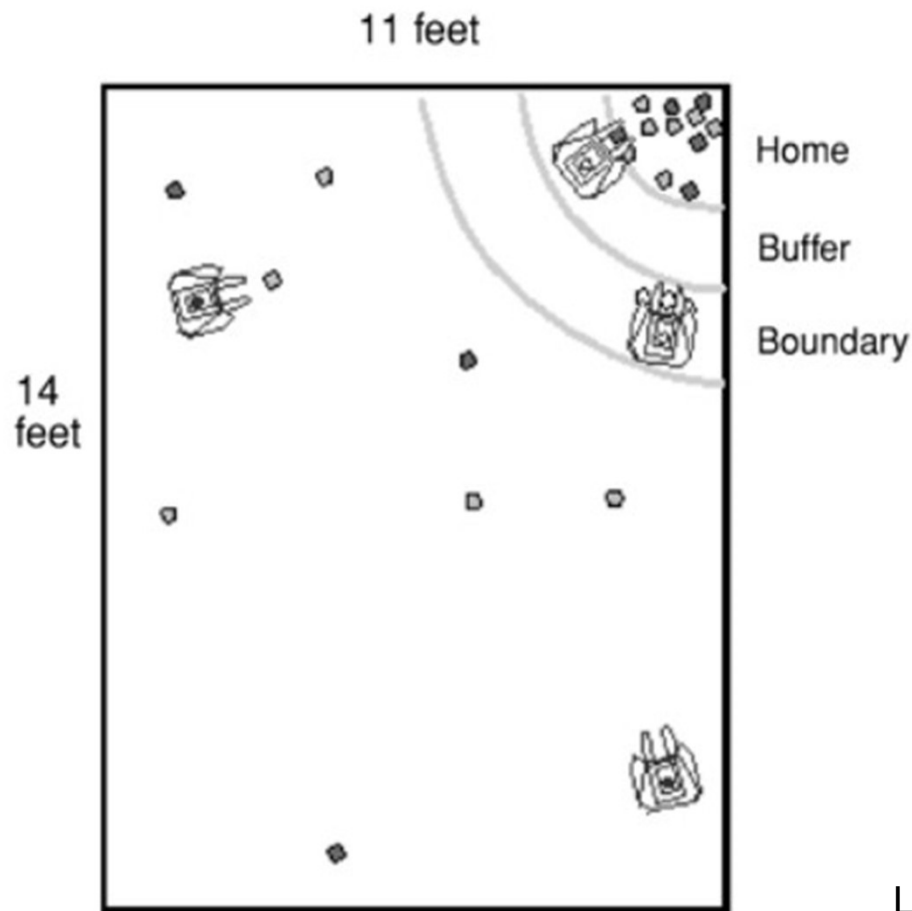
Instead of concentrations of chemical species
we have swarm fractions of robots that are in certain states.

Instead of chemical reactions we have state transitions as effects of robot-robot interactions.

This modeling approach is

- macroscopic
- probabilistic
- non-spatial (well-mixed assumption)

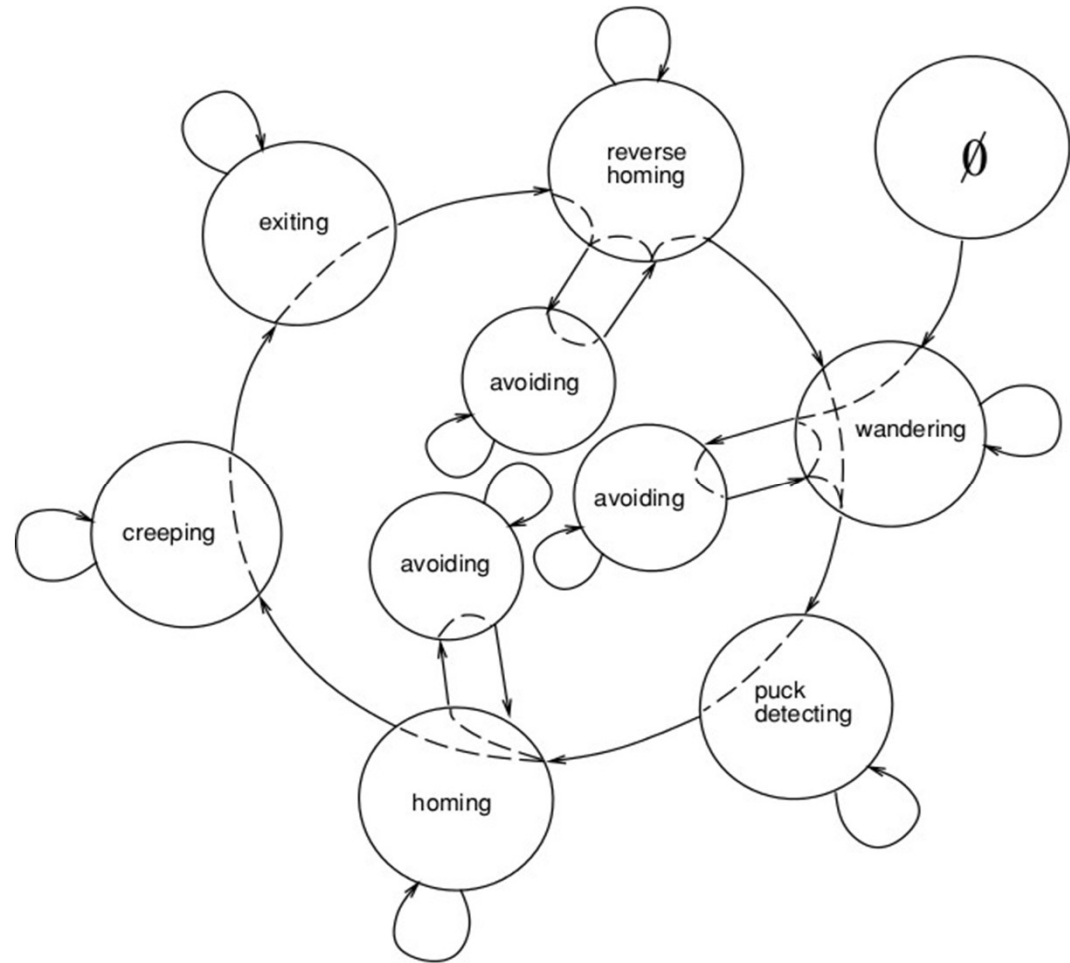
Example: Foraging in a group of robots



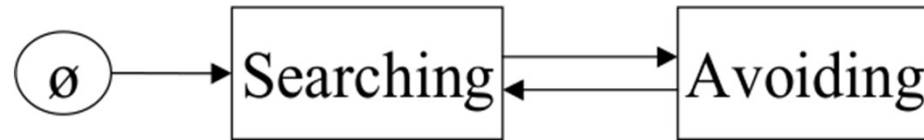
- Avoiding obstacles including other robots
- Wandering or searching for pucks
- Detecting a puck
- Grabbing a puck
- Homing
- Creeping (avoid the boundaries in buffer region)
- Home (robot drops the puck)
- Exiting (robot exits the home region and resumes search)

Lerman and Galstyan (2002)

Full state diagram



A model only of searching and avoiding



$N_s(t)$ is the number of robots in the search state at time t

$N_a(t)$ is the number of robots in the avoiding state at time t

conservation:

$$N_s(t) + N_a(t) = N \text{ total number of robots } N$$

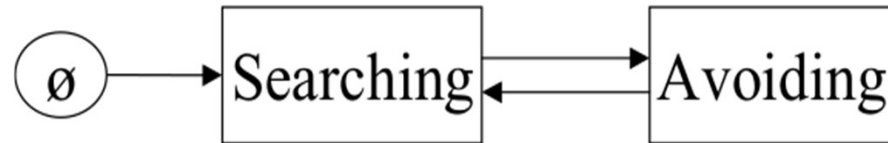
$M(t)$ is the number of uncollected pucks at time t

α_r is the rate coefficient of detecting another robot and

α_p is the rate coefficient of detecting a puck

If a searching robot detects an obstacle, such as another robot or a wall, it executes the avoiding behavior for a time period τ , after which it resumes the search.

Thoughts on how to do this

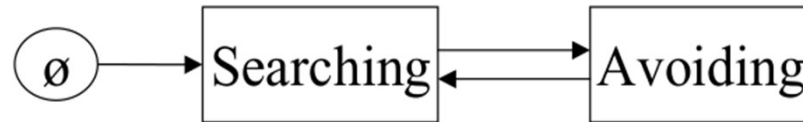


The number of searching robots N_s decreases when two searching robots detect each other and commence avoiding maneuvers or when a searching robot detects another robot in the avoiding state.

N_s increases when robots, that started the avoiding behavior at time $t - \tau$, exit the avoiding state and resume searching.

We don't need an equation describing the dynamics of the avoiding robots N_a , because we can compute this quantity using the conservation of the total number of robots N .

Equations



$$\frac{dN_s(t)}{dt} = -\alpha_r N_s(t)(N_s(t) + N) + \alpha_r N_s(t - \tau)(N_s(t - \tau) + N)$$

$$\frac{dM(t)}{dt} = -\alpha_p N_s(t)M(t)$$

for normalized $n_s = \frac{N_s}{N}$, $n_a = \frac{N_a}{N}$, and $m = \frac{M}{M_{max}}$ we get

$$\frac{dn_s(t)}{dt} = -\alpha_r n_s(t)(n_s(t) + 1) + \alpha_r n_s(t - \tau)(n_s(t - \tau) + 1)$$

$$\frac{dm(t)}{dt} = -\alpha_p n_s(t)m(t)$$

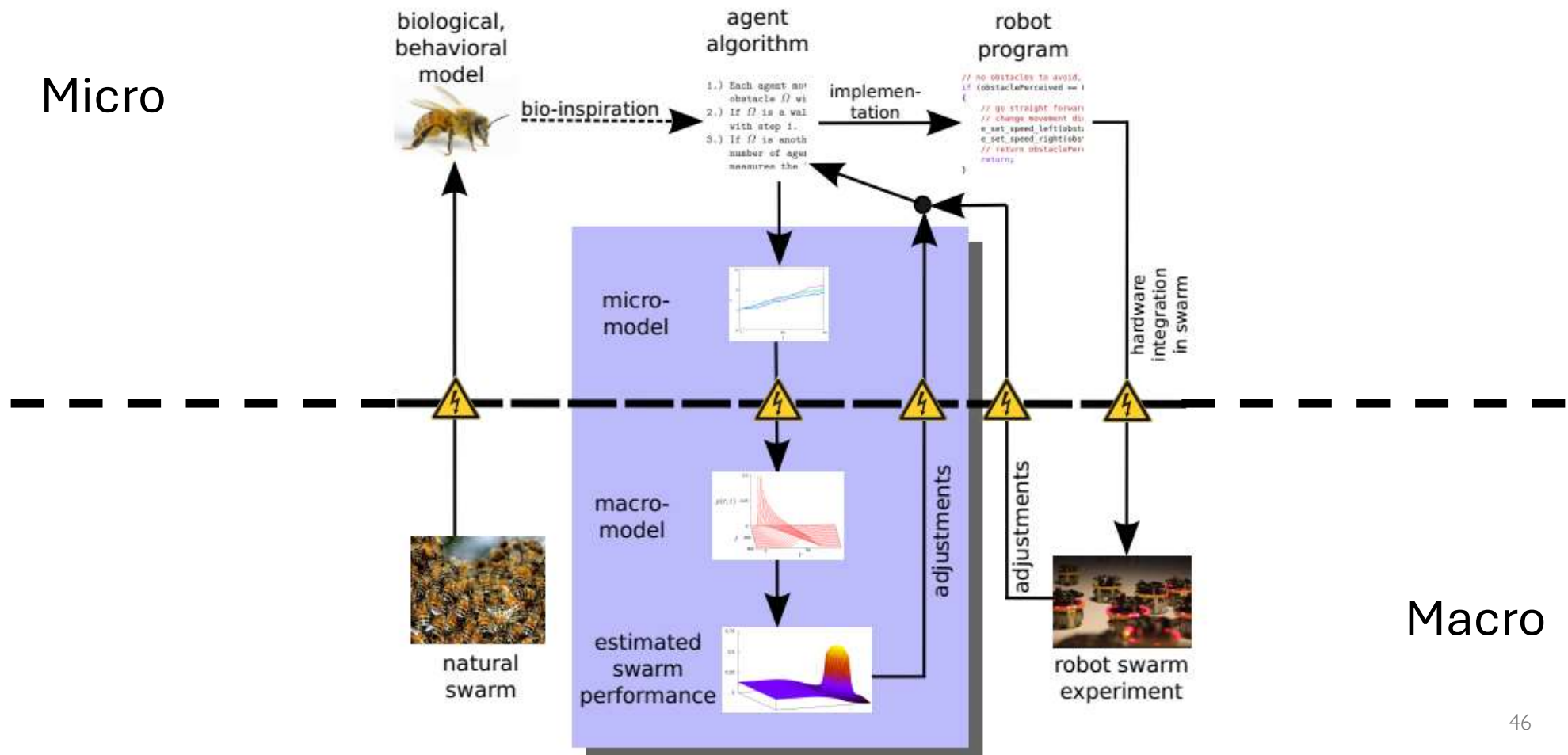
- Why do we use models in swarm robotics instead of relying solely on real-world experiments?
- What is the difference between microscopic and macroscopic modeling in swarm systems?
- Can you name one advantage and one limitation of modeling swarm behaviors using equations like ODEs or PDEs?

Recap



Last
Week
I
Learned...

RECAP: Micro-macro models



RECAP: The rate equations approach

a simple rate equation: $r = kAB$

- chemical reaction between two reactants
- concentrations of the chemical species are A and B
- k is the rate coefficient

Say we have $A + B \rightarrow C$ then an ordinary differential equation (ODE) is defined by the reaction rate:

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for concentrations A, B, and C
(law of mass action)

Micro-macro model based on SDE and PDE

Micro-macro models

idea:

- transition from micro-level to macro-level should be mathematically tractable
- simplification: model a single robot's trajectory (partially) as stochastic process
- justification: locomotion unit of limited accuracy, explicit random behavior (exploration), high number of interactions generates pseudo-randomness

stochastic process

sequence of random variables (time series)

collection of random variables indexed by a set T (time):

$$\{X_t : t \in T\}$$

Towards a microscopic model (1/3)

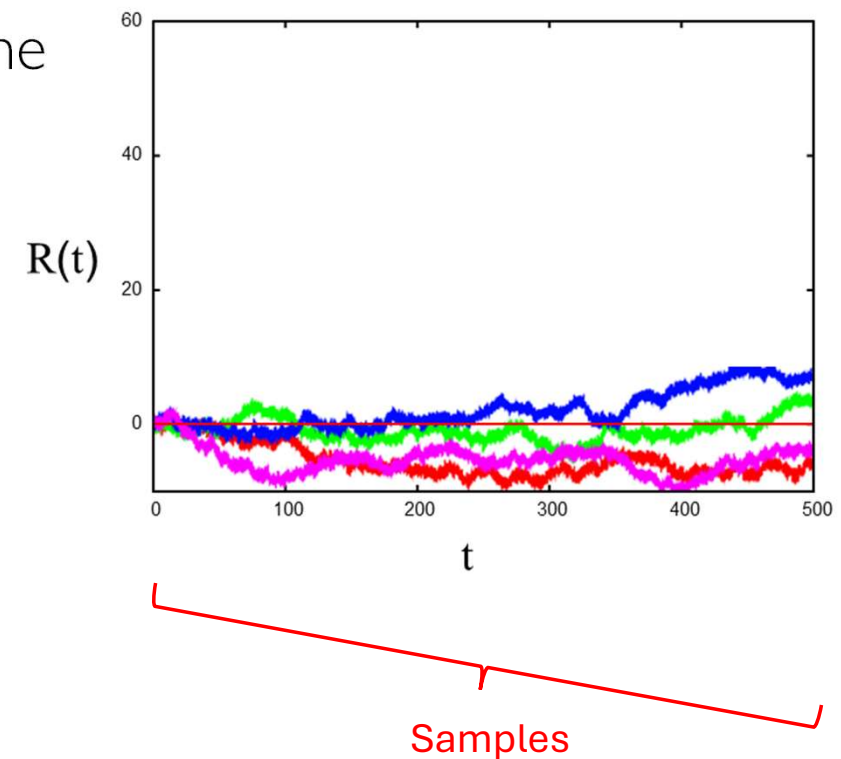
fully stochastic model (random motion) say robot position at time t is $R(t)$ then we can model the robot's trajectory by:

$$\frac{dR(t)}{dt} = \dot{R}(t) = X_t$$

this is a stochastic differential equation (SDE)

X_t is a random displacement of the robot,

X_t is a stochastic process

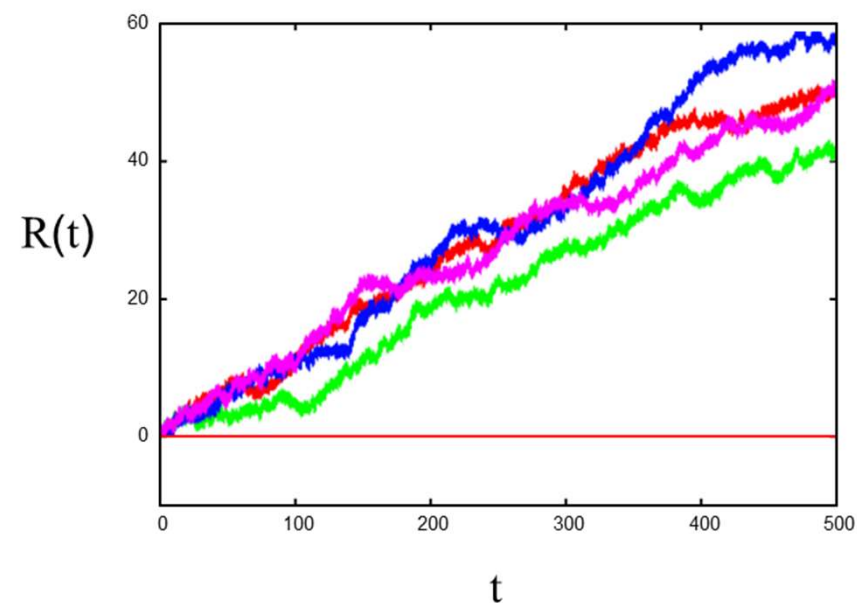


Towards a microscopic model (2/3)

stochastic model with drift

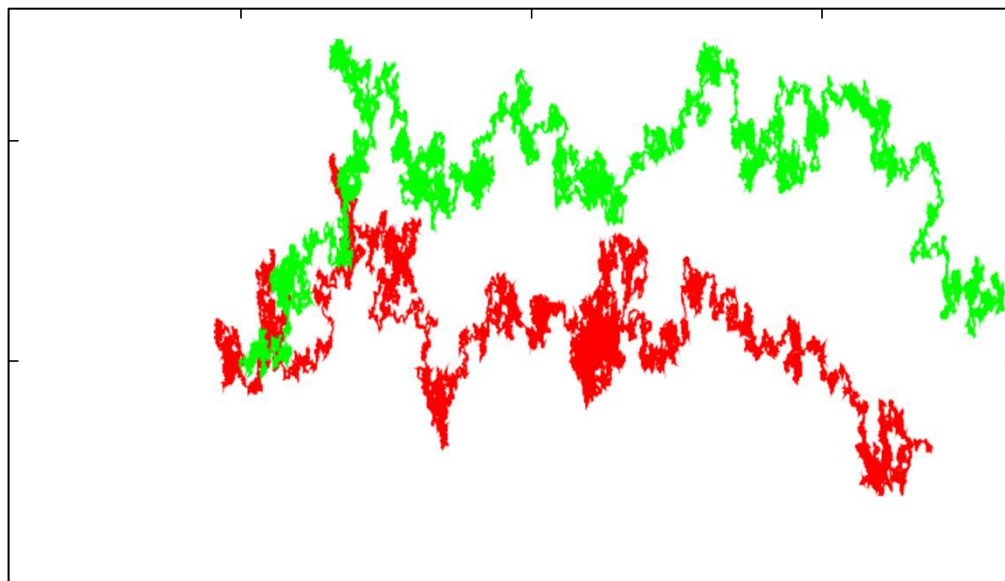
we add a deterministic term, for example,

a constant c : $\dot{R}(t) = X_t + c$



Towards a microscopic model (3/3)

An example for a 2-d stochastic model with drift robot is started in $(0,0)$, two realizations of the stochastic process are shown



Microscopic model

Langevin equation [\(Langevin, 1908\)](#)

Trajectory of a robot given by

$$\dot{R}(t) = \overbrace{A(R(t), t)}^{\text{det.}} + \overbrace{B(R(t), t) F(t)}^{\text{non-det.}},$$

$F(t)$, is stochastic process, e.g. white noise

A describes deterministic behavior

B describes non-deterministic behavior

possible choice of $A(R(t), t) = \nabla P(R(t), t)$

Towards a macroscopic model

- How could a macroscopic model of this look like?
- The **probability density function** of a random variable is a good ‘summary’ of all trajectories that are to be expected.
- We give a different interpretation of a probability density function here as we say it is the **swarm density**. That is the likelihood of encountering a fraction of the swarm at a certain spot.
- How to model the **temporal evolution** of a probability density function? Problem is already solved. Thanks to physicists.

Macroscopic model

Fokker–Planck equation

deducible from Langevin equation

$$\frac{\partial \rho(\mathbf{r}, t)}{\partial t} = -\nabla \cdot (\mathbf{A}(\mathbf{r}, t) \rho(\mathbf{r}, t)) + \frac{1}{2} Q \nabla^2 (B^2(\mathbf{r}, t) \rho(\mathbf{r}, t))$$

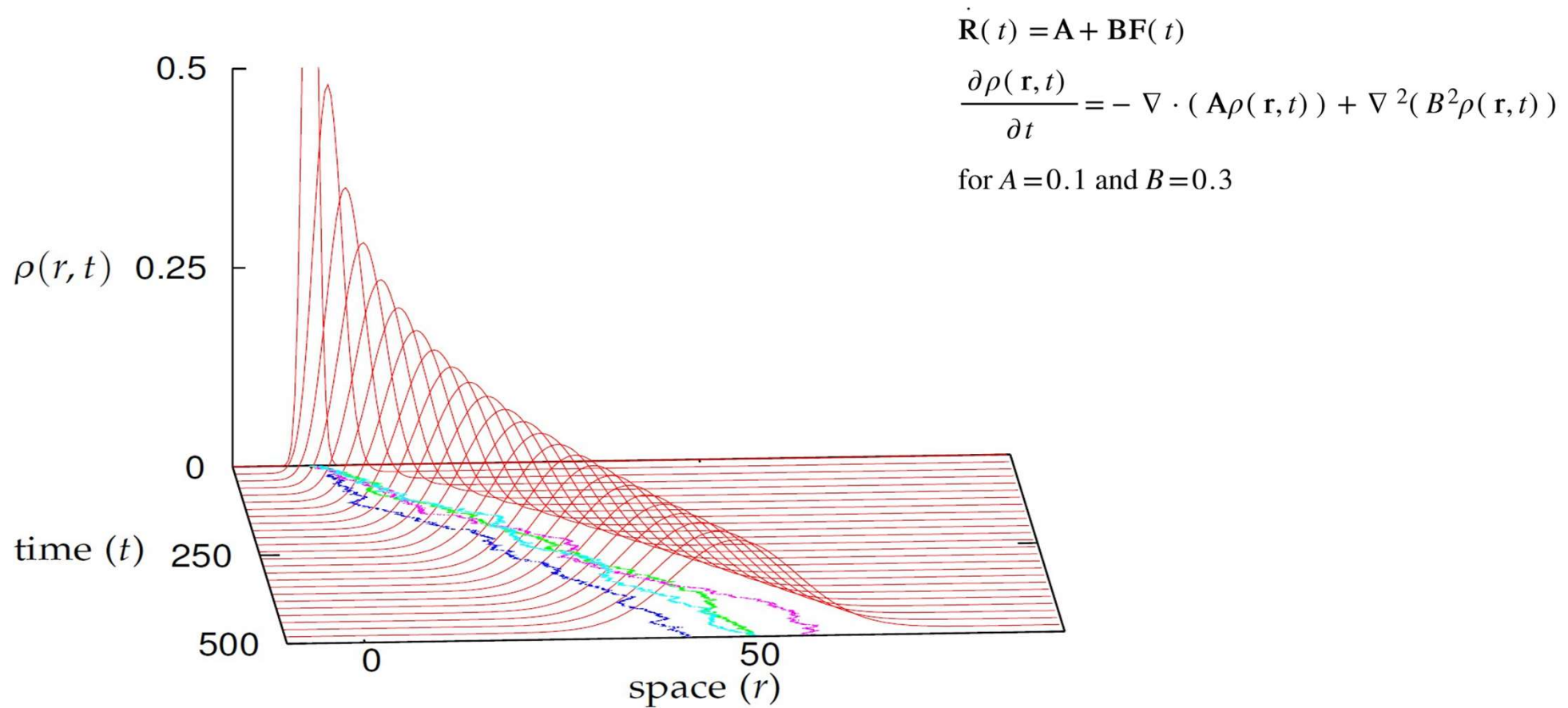
- ρ is probability density for a single robot or (particle) density of all (coexisting) robots at position $\mathbf{r}$ and time t
- ‘nabla’ operator $\nabla = \left(\frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \dots \right)$ is based on the spatial derivatives and gives the gradient
- ∇^2 is based on the 2nd spatial derivatives and is known as the Laplacian operator $\Delta = \nabla \cdot \nabla$
 $= \nabla^2 = \frac{\partial^2}{\partial r_1^2} + \frac{\partial^2}{\partial r_2^2}$

Q results from F (strength of hits)

1st summand: drift term (deterministic behavior)

2nd summand: diffusion term (non-determ. behavior)

Micro-macro model for 1-d space



Random graphs

example of spatial model: random graphs

phase transition

random geometric graphs

Random graphs

An (Erdős–Rényi) random graph is
an undirected graph $\mathbf{G}(\mathbf{X}; \mathbf{p})$,
with a node set $\mathbf{X} = \{0, 1, 2, \dots, N\}$,
each possible edge is included independently with probability p

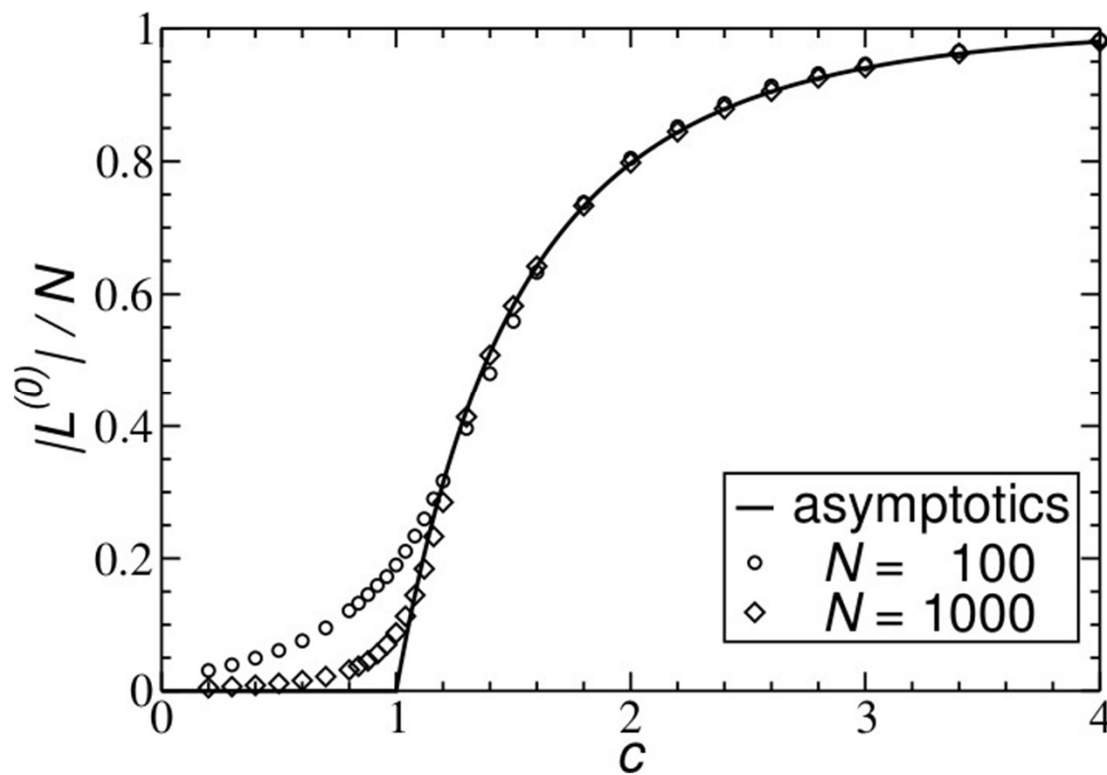
$p = 0$: no edges

$0 < p < 1$: something in between,

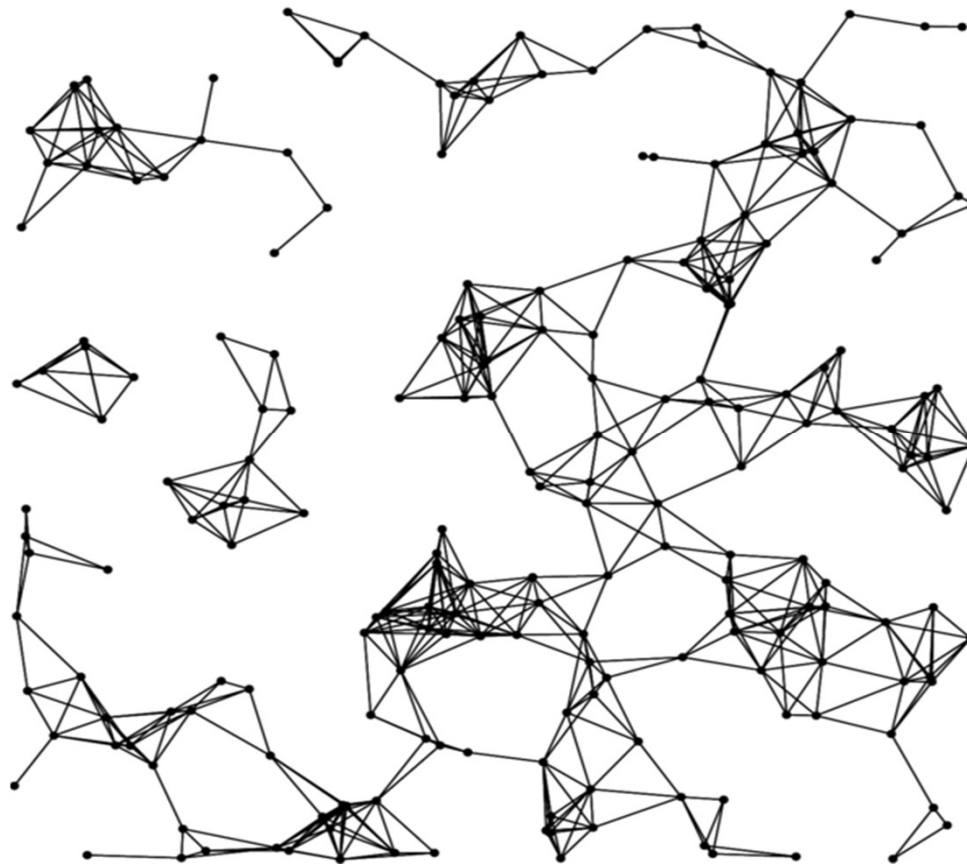
interesting: size of largest connected component

$p = 1$: fully connected graph

Random graphs – Phase transition



Random geometric graphs



Random geometric graphs

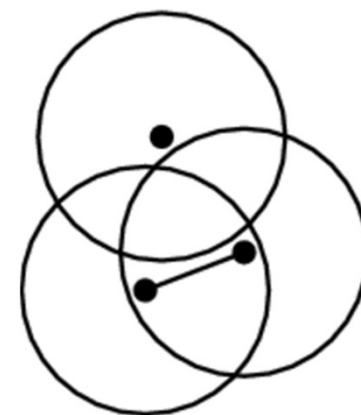
A geometric graph is defined as an undirected graph $G(X; r)$,
with node set $X \subset R^d$,

for 2-d we have $X \subset R^2$,

nodes $x \in X$ and $y \in X$ are connected by an undirected edge
if $\|x - y\| < r$ (unit disc model),

for a norm $\|\cdot\|$ on R^2 (e.g., Euclidean distance)

A random geometric graph is generated by having X sampled
from a probability distribution, for example, uniform distribution
edges not independent as in Erdős–Rényi random graphs



Random geometric graphs and swarm robotics

areas of interest for applications

of random geometric graphs in swarm robotics:

- sizes of connected components (groups of robots that cooperate)
- number of isolated nodes
- (robots without chance to cooperate)
- average degree
- (possibly a measure for robustness)
- When are random geometric graphs a good option to model swarm robotic systems?
- What about dynamic nodes and the dynamics of geometric graphs?

Biological models

real-world model

example: song recognition and localization

phonotaxis robot model

Robots as biological models

robots as model to investigate biological questions and to test hypotheses

instead of

- mere hypothesis
- abstract models (mathematical, computational etc.)
- simulations

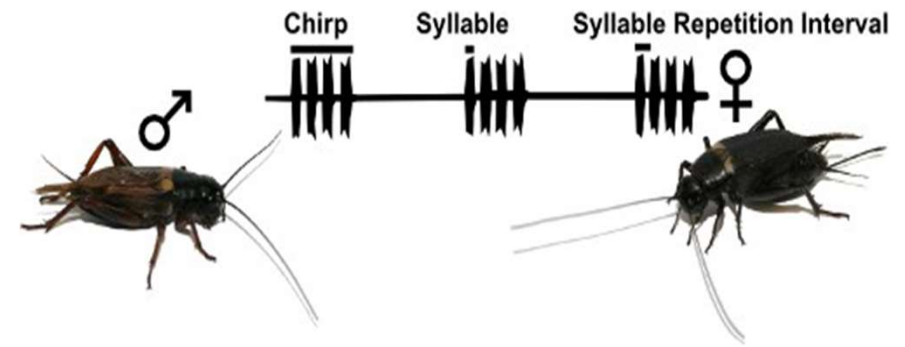


Advantages of a "real-world model"

a robot operating in the real world. . .

- constrains the choice of available models (Webb, 2002)
- forces to be very concrete in specifying the complete biological system helps
- to produce testable hypotheses (Webb, 2001)
- allows to study complex agent–environment interactions

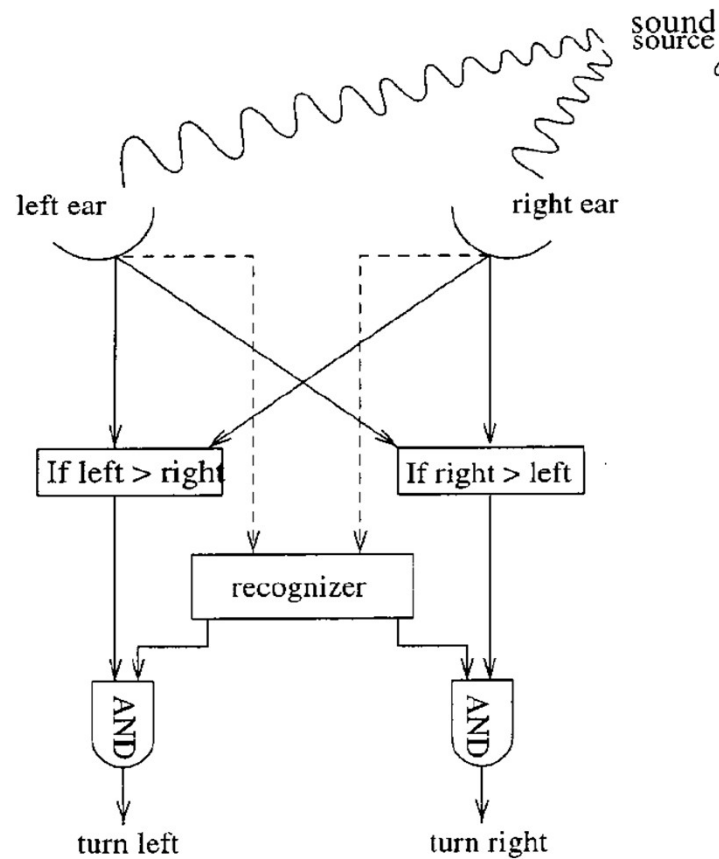
Example: Song recognition and localization



- phonotaxis in crickets
- one ear in each foreleg determines direction-dependent differences
- not just turning towards side with strongest response female crickets recognize males' songs and approach them
- simple model effective due to relative latencies in firing onset and “leaky integration” acting as a low-pass filter, making fast temporal patterns appear continuous

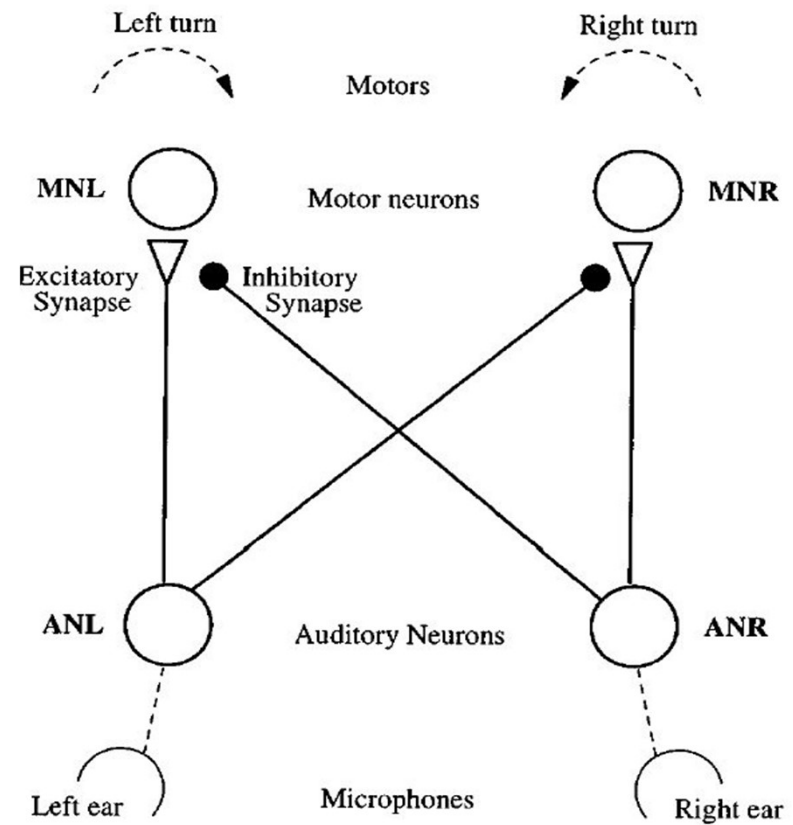
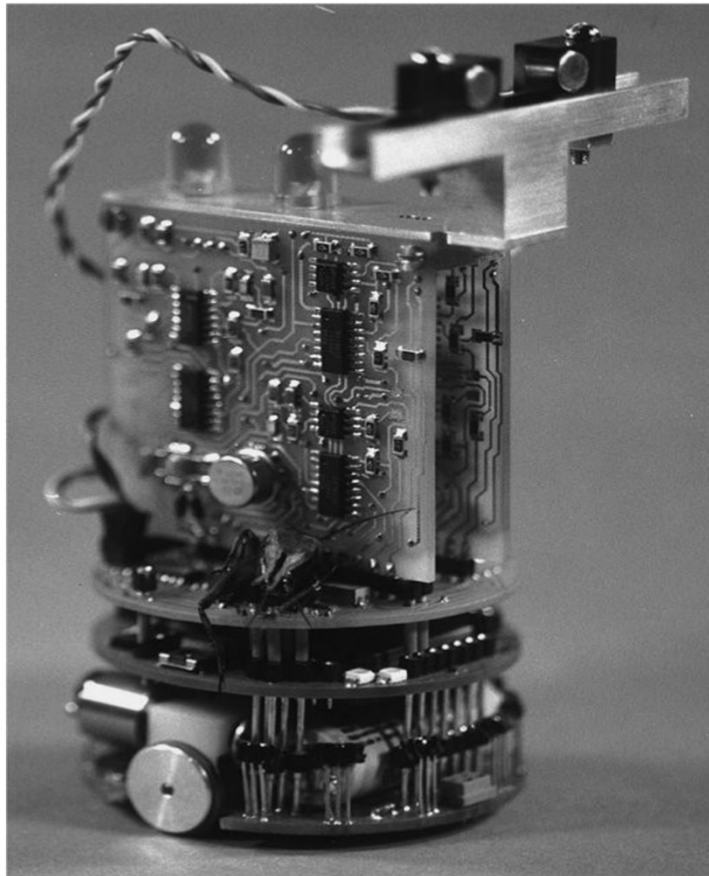
[Webb and Scutt \(2000\)](#)

Phonotaxis robot model: Former, complex model



the signal amplitude at the two ears is compared while the signal pattern is analyzed by low-pass and high-pass filtering

Phonotaxis robot model



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